# Position, distance, displacement

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Distinguish a location, a total path length, and a change in position.

**SUCCESS CRITERIA - I can...**

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

**Core:** teacher selects from 1-9. **Stretch:** 10 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Positions are 15, 10, 5 m at 0, 2, 4 s. Predict position at 7 s. Give the rate, an assumption and a future test.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $\Delta x = x_f - x_i$; distance = sum of path lengths. Choose an origin and positive direction. i = initial; f = final.

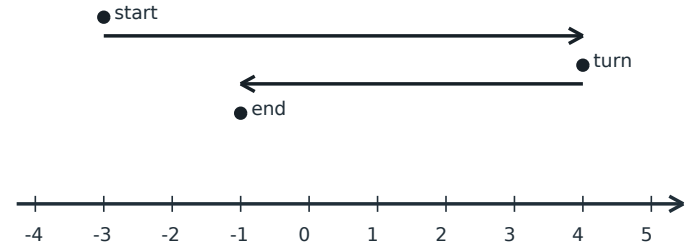
**Worked reasoning:** A walker starts at 2 m, reaches 6 m, then ends at 4 m. Distance=4+2=6 m. Displacement=4-2=2 m right.

**Your turn:** Draw the two route arrows and a separate start-to-finish arrow. Label the length of each.

**Marzano check - circle:** ① ② ③

## 03 Track a signed route

East is positive. Start at -3 m, go to +4 m, then end at -1 m. Sketch the route and calculate distance and displacement.




Position (m); right is positive

**Marzano check - circle:** ① ② ③

## 04 Change the coordinate convention

For the route in 3, reverse the positive direction while keeping the origin fixed. Give the new endpoint coordinates and displacement. What stays unchanged?

**Marzano check - circle:** ① ② ③

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Continue down each column.

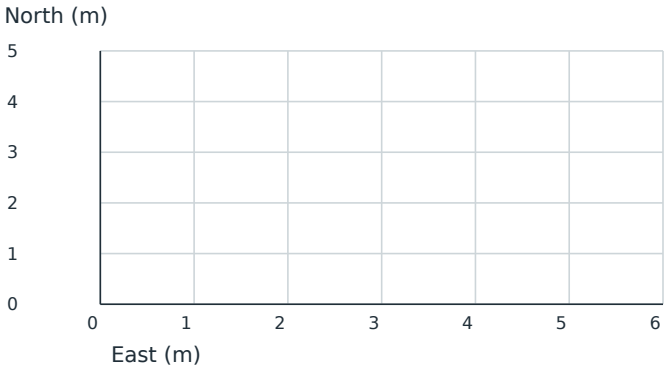
### 05 A closed path

A runner completes one 400 m lap, then runs another 100 m around a circular track. Is the displacement magnitude 500 m? Explain what would be needed to calculate it.

**Marzano check - circle:** 1 2 3

### 06 Different paths, same endpoints

Sketch two different routes from (0, 0) m to (4, 3) m. Give each path distance and the common displacement magnitude.




**Marzano check - circle:** 1 2 3

### 07 Zero and nonzero

Can displacement be zero while distance is nonzero? Can distance be zero while displacement is nonzero? Justify both answers.

**Marzano check - circle:** 1 2 3

### 08 Choose a method

A fictional observing report records 3 images of a moving object but gives no distance or size calibration. State a supported observation and explain why a precise impact prediction is unsupported.

**Marzano check - circle:** 1 2 3

### 09 Exit check

Start at +5 m, walk to -2 m, and finish at +1 m. Calculate distance and displacement. Explain why neither equals final position.

**Marzano check - circle:** 1 2 3

### 10 STRETCH: Apply it another way

In one dimension, start at -2 m and end at +3 m with total distance 11 m. Find two possible single-turn positions and draw both routes.

**Marzano check - circle:** 1 2 3